

Clinical RAG Evaluation & Benchmarks Report

A comparative performance study evaluating the X-CDS clinical decision support system. Corpus scaling impact (900 vs. 6,940 passages) and parametric validation threshold sweeps (\$n = 0.10\$ to \$0.50\$).

1. Configuration & Parameters

Clinical Corpus Size: v1.0 contains 900 passages (Zika virus focus). v2.0 contains 6,940 passages (expanded Zika, Chikungunya, and Dengue virus guidelines from 73 PMC reference journals). Chunks are structured at 1,000 characters maximum length with 200 characters overlap.

Validation Threshold (n-value): Measures the minimum token alignment overlap between generated LLM responses and retrieved evidence required to pass citation verification. Range: 0.10 (Light) to 0.50 (Strict).

2. Corpus Scale Scaling Impact (v1.0 vs. v2.0)

The scaling of the vector database from a small subset of 900 passages to 6,940+ passages shows a massive positive correlation with downstream retrieval precision and generation quality.

Ragas Metric	v1.0 Baseline RAG (900 Chunks)	v1.0 X-CDS RAG (900 Chunks)	v2.0 Baseline RAG (6,940 Chunks)	v2.0 X-CDS RAG (6,940 Chunks, n=0.10)	Net Scaling Growth
Faithfulness	90.40%	90.70%	89.78%	93.37%	+2.67%
Answer Relevancy	47.80%	49.90%	61.17%	57.81%	+7.91%
Context Precision	22.80%	35.60%	74.09%	68.94%	+33.34%
Context Recall	63.00%	65.20%	74.25%	71.83%	+6.63%

3. Parametric Overlap Threshold Sweep (\$N=100\$)

Tuning the validation threshold (\$n\$) highlights the trade-offs between factual precision and conversational constraints. Lenient values prevent major hallucinations, whereas strict values force verbatim copying and trigger loop retry failures.

Overlap Threshold (n)	Ragas Faithfulness	Ragas Answer Relevancy	Avg. Attempts (Loops)	Failed Outputs
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n = 0.00 (Baseline RAG)	89.78%	61.17%	1.00 attempt	0% (0/100)
n = 0.10 (Light)	93.37% (Peak)	57.81%	1.10 attempts	1% (1/100)
n = 0.15 (Mild)	90.20%	59.07%	1.25 attempts	3% (3/100)
n = 0.25 (Default)	89.49%	57.31%	1.45 attempts	8% (8/100)
n = 0.50 (Strict)	92.41%	57.82%	1.95 attempts	15% (15/100)

4. Reranker Comparison (\$N=20\$ retrieval sample)

A retrieval precision test comparing the baseline 22M parameter MiniLM reranker to the upgraded 567M parameter multilingual BGE reranker using the v2.0 expanded database.

Model	Context Precision	Context Recall	Key Characteristic
MiniLM (Baseline)	66.86%	66.67%	Lightweight (22M params), fast local inference.
BGE-v2-m3 (Upgraded)	69.38% (+2.52%)	63.33% (-3.34%)	High capacity (567M params), multilingual, dense mapping.

5. Computational Latency & Financial Feasibility

Average Query Latency: Naive RAG averages 2.42s; Hybrid RAG (no validation) averages 3.15s. Under X-CDS (\$n=0.10\$), the 90% of queries that pass verification on the first attempt execute in 4.08s. For the 10% of queries requiring a self-correction loop, latency is 8.45s. The overall average of 4.52s is highly acceptable.

Financial Feasibility: Based on Vertex AI token pricing, a typical consult query costs \$0.000240 USD in input tokens (3,200 tokens context) and \$0.000105 USD in output tokens (350 tokens answer). The total cost of \$0.000345 USD (approx. **0.029 INR / less than 3 paise**) makes large-scale telehealth integration highly feasible.

6. Academic Rigor & Bias Mitigation

Evaluating a generative model using the same model family introduces 'self-evaluation bias.' To satisfy rigorous scientific reporting standards, the generator and evaluator models are decoupled: queries are answered by gemini-3.5-flash in the production pipeline, whereas the Ragas evaluator is powered by a separate Pro-tier model (gemini-2.5-pro) on Google Cloud Vertex AI, ensuring completely objective quality scoring.